

TMS

Tuition Management System

UI PRD — Consolidated Design Specification

Strategy & Personas · Design System · Module 01 · D9 Smart Admin Dashboard

Document Title	TMS UI PRD — Consolidated Design Specification
Covers	Strategy & Personas · Design System · Module 01 (Auth & Onboarding) · D9 (Smart Admin Dashboard)
Related Agreement	SOW-001, Phase 1: Core Foundation — Lucaz Soft Pvt. Ltd. & Sanskardip Shikshalaya
Governing PRD	PRD v2.0 dated 10 May 2026 (Annexure A to MSA, 29 June 2026)
Version	v2.0
Date	7 July 2026
Prepared By	UI/UX & Frontend Design — Lucaz Soft Pvt. Ltd.
Status	For Client Review

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1. Project Overview

TMS is a multi-tenant, multi-branch tuition and school management platform being built for Sanskardip Shikshalaya (Damak, Jhapa) under a phased engagement. Phase 1 establishes the foundation: architecture, authentication/RBAC, tenant & branch management, student enrollment, timetabling, geo-attendance, fee billing, and role-scoped dashboards. This section defines the user-centered strategy behind the screens specified later in this document — who they are for, what those users need, and why key decisions were made.

1.1 Business Goals

- Digitize enrollment, attendance, and billing for a multi-branch institution currently reliant on manual/paper processes.
- Reduce administrative overhead by giving Tenant and Branch Admins a single, consolidated operational view.
- Reduce fee-collection leakage through automated dues tracking and access-blocking, with a clear admin override path.
- Build a white-label foundation that Lucaz Soft can extend to additional institutions beyond this engagement.
- Deliver value in verifiable phases — Phase 1's 62-point acceptance checklist exists specifically so trust is established incrementally, not promised upfront.

1.2 User Goals

- Admins want visibility across every branch without visiting each one — one dashboard, real numbers.
- Teachers want attendance and daily logging to take seconds, not minutes, and to work reliably even with a basic Android phone and inconsistent signal.
- Parents want to know, at a glance, whether their child attended class and whether fees are paid — without navigating a complex system.
- Students want their own record (timetable, fees, ID) in one place.
- Every role wants confidence that they only ever see what belongs to them — no cross-tenant or cross-role data leakage.

1.3 Target Audience

Primary audience: staff and families of small-to-mid-sized Nepali tuition centers and schools, starting with a single institution running three branches. This audience spans a wide technical comfort range — from administrators comfortable with spreadsheets and dashboards, to teachers and parents whose primary computing device is an entry-to-mid-range Android phone on a mobile data connection. This constraint shapes nearly every design decision in this document: SMS is treated as a first-class notification channel (not a fallback), mobile flows are optimized for one-handed, low-bandwidth use, and admin-only complexity (RBAC, multi-tenant settings) is kept out of teacher/parent/student surfaces entirely.

1.4 User Personas

Deepa Yadav — Tenant Admin / Director

Role	Owns institutional outcomes across all branches; final approver on staffing, finance, and policy.
Quote	"I need to know how every branch is doing without calling three branch admins every morning."
Goals	Consolidated multi-branch visibility; confidence that fee collection and attendance are on track; fast approval flows for escalated leave requests.

Pain Points	Currently reconciles branch reports manually over phone calls and WhatsApp; no single source of truth; delayed visibility into blocked/at-risk students.
Primary Device	Laptop (primary), phone (secondary)
Tech Comfort	Comfortable with business software; limited patience for unnecessary complexity.

Branch Admin — Damak Branch Coordinator

Role	Runs day-to-day operations for one branch: staffing, attendance, fee follow-up, timetabling.
Quote	"I need to know, right now, who's absent, who's blocked, and what still needs my sign-off."
Goals	Same-day operational clarity; quick override tools with an audit trail; a timetable that flags conflicts before they become a problem.
Pain Points	Currently tracks attendance and dues in separate paper registers and spreadsheets; conflicts (double-booked teachers/rooms) are caught late.
Primary Device	Laptop/desktop at the branch, phone for on-the-go checks
Tech Comfort	Moderately comfortable; prefers clear labels over icons alone.

Ramesh Koirala — Teacher

Role	Teaches Mathematics across two branches; paid partly on attendance-linked metrics.
Quote	"I just want to walk in, tap once, and get back to teaching — not fight with an app between classes."
Goals	One-tap, reliable geo-attendance; clear session status so there's no dispute later; minimal typing.
Pain Points	Inconsistent GPS/network near classrooms; anxiety about being wrongly marked absent due to a technical glitch; daily update logging feels like an extra chore.
Primary Device	Personal Android phone (mobile app)
Tech Comfort	Basic-to-moderate; needs large tap targets and immediate, unambiguous feedback.

Priya Sharma — Student

Role	Grade 9 student enrolled in Regular Class plus a personalized Music class.
Quote	"I want to see my timetable and fee status without asking my parents to check for me."
Goals	Quick access to today's schedule, fee status, and her Digital ID; minimal navigation depth.
Pain Points	Limited screen time allowed by parents; needs the essentials fast, not a feature-rich app to explore.
Primary Device	Shared family phone (mobile app), occasionally a laptop
Tech Comfort	Comfortable with apps generally; low tolerance for slow loading or clutter.

Sarita Sharma — Parent / Guardian

Role	Mother of two children enrolled at different grade levels in the same branch.
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Quote	"I need to know both my kids are fine and I'm not missing a fee deadline — in one place, in a language I can act on immediately."
Goals	A child switcher so she isn't logging in twice; fee reminders that state the total payable clearly; SMS as a reliable backup to push notifications.
Pain Points	Juggles two children's schedules and fees separately today; worried about missing a due date and having a child blocked from class unexpectedly.
Primary Device	Personal Android phone (mobile app)
Tech Comfort	Basic; strongly prefers SMS/plain notifications over needing to open and navigate an app.

2. User Flow

Core flows delivered in Phase 1, shown as the sequence of screens/decisions a user moves through. Each flow was validated against the acceptance criteria in SOW-001 Section 7.

2.1 Authentication Flow

- Enter email + password → validate → on success: mandatory 2FA for Super Admin/Tenant Admin → OTP verified → role-based dashboard redirect.
- On failure: inline error with a live attempt counter → 5th failure locks the account and offers Reset Password or Contact Admin.

2.2 Password Recovery Flow

- Forgot Password → enter registered email → 6-digit OTP sent (SMS + email) → OTP verified (auto-submits on the 6th digit) → set new password against 5 live validation rules → redirected to Login with a success confirmation.

2.3 Teacher Geo-Attendance Flow

- Teacher opens the mobile app near class time → app checks GPS against the branch's configured radius → Mark IN enabled only inside the radius → session begins.
- If the teacher leaves the radius without tapping Mark OUT, auto-departure fires after the grace period, timestamped to the last known in-radius position, and immediately notifies the Branch Admin.
- Returning and tapping Mark IN again resumes the session; total in-premises time is calculated from all stamps rather than a single in/out pair – this protects teachers from being penalized for brief, legitimate exits.

2.4 Fee Payment Flow

- Invoice generated (pro-rated automatically from admission date) → combined + per-course invoice created → SMS sent to parent with the total payable → parent scans the Nepal Pay QR code → payment webhook marks the invoice Paid and confirms via SMS + push, with manual confirmation available as a fallback.

2.5 Parent Oversight Flow

- Parent logs in → selects a child via the child switcher (if more than one is linked) → dashboard scopes every widget (attendance, fees, timetable, announcements) to that child → switching children never requires a separate login.

3. Information Architecture

Each role's navigation is scoped to only what that role can act on — there is no shared admin surface that a lower-privilege role can see into, even in a read-only or greyed-out form (with the deliberate exception of Phase 2/3 feature previews, discussed in Section 9).

Role	Primary Sections
Tenant Admin	Dashboard, Branches, Students, Teachers, Timetables, Courses, Fee & Billing, Leave Management, Resource Logs, Tenant Settings, RBAC & Roles
Branch Admin	Dashboard, Students, Teachers, Timetable, Attendance, Personalized Classes, Fee & Billing, Leave Requests, Resource Logs
Teacher	Today's Classes, My Timetable, Attendance, Daily Update Log, My Profile, Leave Requests
Student	Home, Timetable, Fees, Digital ID
Parent	Home (with child switcher), Attendance, Fees

3.1 Sitemap (Phase 1 routes)

Route	Page	Role Access	Auth Required
/login	Login Page	All	No
/forgot-password	Forgot Password	All	No
/reset-password/:token	Reset Password	All	No
/tenant/dashboard	Tenant Admin Dashboard	Tenant Admin	Yes
/tenant/branches	Branch Management	Tenant Admin	Yes
/branch/dashboard	Branch Admin Dashboard	Branch Admin	Yes
/branch/students	Student Management	Branch Admin	Yes
/branch/attendance	Attendance Overview	Branch Admin	Yes
/branch/fees	Fee & Billing	Branch Admin, Accountant	Yes
/teacher/dashboard	Teacher Dashboard	Teacher	Yes
/teacher/attendance/:classId	Mark Attendance	Teacher	Yes
/student/home	Student Home	Student	Yes
/parent/home	Parent Home	Parent	Yes

Full route map, including Phase 2–4 routes, is maintained in the engineering routing table — this sitemap reflects Phase 1 scope only.

4. Wireframe Structure

Given the well-defined component specification already provided by the client's PRD and SOW, low-fidelity wireframes were compressed into structural decisions validated directly in high-fidelity interactive prototypes — avoiding a throwaway mid-fidelity stage for a scope this well-specified.

4.1 Layout Patterns Used

- Split-screen authentication layout: a narrative/brand panel paired with a focused single-task form panel — used for Login, where establishing trust matters as much as the task.
- Centered card layout: a single focused task with no distractions — used for Forgot Password, OTP Verification, Reset Password, and 2FA.
- Sidebar + content shell: persistent collapsible navigation with a scrollable widget grid — used for every dashboard, so switching sections never reloads the whole shell.
- Card-grid content hierarchy inside dashboards: KPIs (glanceable numbers) sit above lists (actionable detail), which sit above previews (Phase 2/3 context).

4.2 Delivered Prototypes

File	Fidelity	Covers
tms-auth-prototype.html	High-fidelity, interactive	Login, Forgot Password, OTP, Reset Password, 2FA
tms-dashboards-prototype.html	High-fidelity, interactive	Tenant Admin, Branch Admin, Teacher, Student, Parent dashboards

5. Design System

This section defines the shared visual language applied consistently across both modules covered in this document. All colors, type, and component states below are the single source of truth for engineering handoff.

5.1 Brand Color Palette

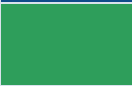
Primary brand palette, as supplied by the client and applied across all UI surfaces.

Swatch	Color Name	Hex Code	RGB	Usage
	Deep Academic Blue	#0F4C8A	15, 76, 138	Primary brand color · page headings · sidebar navigation background
	Royal Blue	#1B5FA7	27, 95, 167	Secondary buttons · links · focus states · active nav accents
	Golden Orange	#F39C12	243, 156, 18	Primary CTA buttons (Sign in, Send OTP, Verify) · highlights · KPI icons
	Bright Gold	#F7B733	247, 183, 51	Hover state for primary buttons · decorative accents · seal motif
	Pure White	#FFFFFF	255, 255, 255	Page and card backgrounds · form panels
	Light Gray	#F5F7FA	245, 247, 250	Section backgrounds · zebra table rows · app canvas
	Dark Slate	#2C3E50	44, 62, 80	Primary body text · headings on light surfaces

Usage column reflects actual application in the delivered prototypes, not just intended use.

5.2 Functional / Status Colors

Status colors are used strictly for feedback (validation, state, outcome) and are kept separate from brand colors so meaning is never ambiguous.

Swatch	Color Name	Hex Code	RGB	Usage
	Success	#2E9E5B	46, 158, 91	Paid / confirmed / approved states · strong password
	Warning	#E08E00	224, 142, 0	Pending items · fair password strength · grace-period notices
	Error	#D64545	214, 69, 69	Invalid input · blocked students · failed validation
	Info	#1B5FA7	27, 95, 167	Neutral / upcoming status tags · informational badges

5.3 Typography

Two type families are paired deliberately: Fraunces, a characterful serif, carries headlines, page titles, and large numeric data — giving the product an academic, trustworthy tone appropriate to an educational institution. Roboto (matching the Material UI v5 foundation specified for engineering) handles all UI text.

Role	Typeface	Size / Weight	Applied To
Hero headline	Fraunces (Display)	34px / 600	Login hero panel — "Welcome to TMS"

Role	Typeface	Size / Weight	Applied To
Page title (H2)	Fraunces (Display)	22–26px / 600	"Sign in", "Set new password", dashboard section titles
Section label (H3)	Roboto	14.5px / 700	Card headers — "Branch-wise Summary", "Today's Timetable"
Body text	Roboto	14–15px / 400	Descriptions, helper text, table content
Label / caption	Roboto	11.5–12.5px / 500	Field labels, badges, timestamps, hints
Button label	Roboto	13–15px / 700	All buttons and CTAs
Data / numeric (KPI)	Fraunces (Display)	22–28px / 600	KPI card figures — "847", "94%"

5.4 Spacing Scale

Value	Applied To
4px	Icon-to-text gaps, tight inline spacing
8–10px	Field label to input, badge padding
14–18px	Between stacked rows within a card
18–22px	Gap between cards in a grid
26–30px	Card internal padding, section separation
44–56px	Panel padding on large surfaces (login hero/form panels)

5.5 Radius & Elevation

Surface	Radius	Shadow
Buttons / inputs	7–8px radius	No shadow at rest; primary button gets a soft gold glow shadow
Cards / widgets	10–14px radius	0 10px 28px -14px rgba(15,76,138,0.18) — soft, cool-toned elevation
Modals / center-cards	14–20px radius	Stronger elevation: 0 12px 32px -12px rgba(15,76,138,0.22)
Pills / badges	20px radius (full)	No shadow — flat fill with tinted background + colored text

5.6 Responsive Breakpoints

Breakpoint	Width	Layout Behavior
xs (Mobile)	0–599px	Single column. Temporary drawer. Bottom nav (mobile app).
sm (Tablet Portrait)	600–959px	Single/two column. Temporary drawer.
md (Tablet Landscape)	960–1279px	Two column. Temporary drawer.
lg (Desktop)	1280–1919px	Full layout. Persistent collapsible sidebar.
xl (Wide Desktop)	1920px+	Full layout. Persistent sidebar. Wider content area.

5.7 Core Component Reference

Component states referenced throughout Sections 6–7. Engineering should treat this as the canonical state list per component.

Component	States	Notes
Primary Button	Default · Hover · Loading (spinner) · Disabled	Golden Orange fill, Bright Gold on hover, muted tan when disabled
Secondary Button	Default · Hover	Transparent with Royal Blue border/text; light blue tint on hover
Text Input (standard)	Default · Focus · Error	Underline style; Royal Blue on focus, Error red with helper text on invalid
OTP Input (6-cell)	Empty · Filled · Focus	Auto-advance and auto-submit on 6th digit; Royal Blue border when filled
Checkbox / Switch	Unchecked · Checked	Royal Blue accent color
Snackbar	Error · Success	Dark slate background with a 4px colored left border
Dialog / Modal	Default	White surface, centered, strong elevation — account-lock and session timeout
KPI Card	Loading (skeleton) · Loaded	Shimmer skeleton placeholder before live data resolves (D9.6)
Progress Ring	Animated fill	SVG stroke animates from 0 to target % on screen load
Status Badge / Pill	Success · Warning · Error · Info · Gold	Tinted background + matching colored text, 20px full radius
Sidebar Nav Item	Default · Hover · Active · Disabled ("Soon")	Active item gets a Golden Orange left border + tinted background; Phase 2/3 items shown disabled with a phase tag

6. Module 01 — Authentication & Onboarding

The entry point for all user roles across web and mobile. Uses email + password authentication with role-aware redirection, mandatory 2FA for admin roles, and a full password-recovery flow. Delivered as an interactive HTML prototype: tms-auth-prototype.html.

6.1 Login Page (01.1)

Split-screen layout on desktop: a brand storytelling panel (Deep Academic Blue gradient, feature highlights, institutional seal motif) on the left, and the sign-in form on a white panel to the right. Collapses to a single column below 980px.

Component Specification

Component	Type / Props	Validation	Behavior
Email Input	Standard TextField, type=email, required	Valid email (RFC 5322); cannot be empty	Inline error on blur if invalid
Password Input	Standard TextField, type=password, required	Min 1 character on login (no strength check here)	Show/Hide toggle via eye icon
Remember Me	Checkbox	—	Stores auth token 30 days if checked; session-only otherwise
Forgot Password Link	Text link	—	Navigates to Forgot Password screen
Sign In Button	Contained, primary (Golden Orange)	Disabled until both fields are non-empty	Shows spinner on submit; disabled while loading
Error Snackbar	Severity: error, 4s auto-hide	—	Shows attempt counter: "3 attempts remaining before account lock"
Account Lock Dialog	Modal, 2 actions	—	Triggered after 5 failed attempts; offers Reset Password or Contact Admin

Post-Login Role Redirection

Role	Redirect Target
Super Admin	/super-admin/dashboard
Tenant Admin	/tenant/dashboard
Branch Admin	/branch/dashboard
Teacher	/teacher/dashboard
Accountant	/staff/finance
Receptionist	/staff/reception
Janitor / Cleaner	/staff/tasks
Student	/student/home
Parent / Guardian	/parent/home

6.2 Forgot Password & OTP Verification (01.2)

Two-step recovery flow: the user enters their registered email, receives a 6-digit OTP by SMS/email, and enters it to proceed to Reset Password. Centered card layout, consistent across both steps.

Component	Type / Props	Validation	Behavior
Email Input	Standard TextField, required	Valid email, must match a registered account	Triggers OTP generation on submit
Send OTP Button	Contained, primary	Disabled if empty or invalid format	Transitions to OTP step on success
OTP Input (×6)	Individual digit fields, numeric only	All 6 must be filled	Auto-focus next field; auto-submit on 6th digit
OTP Timer	Countdown, 5:00 default	—	Shows "Resend OTP" once expired
Resend OTP	Text link, disabled until expiry	—	Re-triggers OTP generation and resets timer
Verify OTP Button	Contained, primary	Disabled until all 6 digits entered	On success, routes to Reset Password

6.3 Reset Password (01.3)

New password must satisfy five rules, all shown as a live checklist beneath the field. A strength meter gives immediate visual feedback.

Validation Rules

Rule	Validation	Error Message
Minimum Length	≥ 8 characters	Password must be at least 8 characters
Uppercase	≥ 1 uppercase letter (A–Z)	Must include at least one uppercase letter
Lowercase	≥ 1 lowercase letter (a–z)	Must include at least one lowercase letter
Number	≥ 1 digit (0–9)	Must include at least one number
Special Character	≥ 1 of !@#\$%^&*	Must include at least one special character
Confirm Match	Must equal the new password exactly	Passwords do not match
Not Previous	Cannot match current/last 3 passwords (server-side)	Cannot reuse a recent password

Password Strength Indicator

Score	Label	Bar Color	Criteria
1/4	Weak	Error red	Only 1–2 rules met
2/4	Fair	Warning amber	3 rules met
3/4	Good	Success-adjacent teal-green	4 rules met
4/4	Strong	Success green	All 5 rules met

6.4 Two-Factor Authentication (01.6)

Mandatory for Super Admin and Tenant Admin; optional (Tenant-Admin-enabled) for Branch Admin. Triggered on login from an unrecognized device or location.

Rule	Detail
Trigger	New device, new IP location, or admin-enforced (Super Admin / Tenant Admin: always)
Method	SMS OTP to registered phone (primary); Email OTP as fallback

Rule	Detail
Code Length	6 digits, numeric only
Expiry	5 minutes; admin-configurable 3–10 min range
Max Attempts	3 attempts before the code is invalidated and a new one must be requested
Trust Device	30-day cookie when the checkbox is selected; bypasses 2FA on that device for 30 days
Roles Required	Super Admin: always. Tenant Admin: always. Branch Admin: optional, Tenant-Admin-enabled.

7. Smart Admin Dashboard (D9) – Phase 1

Real-time, role-scoped dashboards for Tenant Admin, Branch Admin, Teacher, Student, and Parent. Delivered as an interactive HTML prototype: [tms-dashboards-prototype.html](#). Each role sees only the data it is authorized to view (D9.7).

7.1 Tenant Admin Dashboard

Consolidated view across all branches of Sanskardip Shikshalaya (Damak, Birtamod, and Charali).

Widget	Shows	Ref
KPI: Total Students	Count across all branches, with month-over-month delta	D9.1
KPI: Active Teachers	Count across all branches, with new-hires-this-month delta	D9.1
KPI: Collection vs Target (MTD)	Progress bar, % of monthly fee-collection target reached	D9.1
KPI: Pending Alerts	Count of items needing Tenant Admin attention	D9.1
Branch-wise Summary	Horizontal comparison bars — fee collection % per branch	D9.1
Pending Alerts Feed	Tagged list: Fee / Attendance / Setup / Leave, each with context	D9.1

7.2 Branch Admin Dashboard

Scoped to a single branch's day-to-day operations.

Widget	Shows	Ref
Today's Teacher Attendance	Animated progress ring, % present today	D9.2
Today's Student Attendance	Animated progress ring, % present today	D9.2
Blocked Students	Count of fee-blocked students, with today's override count	D9.2
Pending Fee Invoices	Count plus outstanding amount in NPR	D9.2
Today's Timetable	Time-ordered list: class, room, teacher, session status badge	D9.2
Resource Log Status	Checklist of daily logs with Logged / Pending / Overdue status dots	D9.2

7.3 Teacher Dashboard

Focused on the teacher's own classes and administrative obligations for the day.

Widget	Shows	Ref
My Classes Today	Time-ordered list with room, enrollment count, and session status	D9.3
Pending Daily Update Log	Outstanding per-class submissions with a one-click submit action	D9.3
Recent Homework Submissions	Preview widget, clearly tagged as a Phase 2 feature preview	D9.3

7.4 Student Dashboard

Widget	Shows	Ref
Today's Timetable	Time-ordered list with room, teacher, and attendance status	D9.4
Fee Status	Current standing (Paid / Due) plus next invoice date	D9.4
Announcements	Branch-level notices, most recent first	D9.4
Digital Student ID	ID summary card with a Download PDF action	D9.4 / D10

7.5 Parent Dashboard

Includes a multi-child switcher; all other widgets scope to the currently selected child.

Widget	Shows	Ref
Child Switcher	Chip selector across all linked children; all widgets below scope to the selected child	Parent PRD note
Today's Attendance	Present / Absent / Excused status for the selected child	D9.5
Pending Fee Balance	Outstanding amount in NPR for the selected child	D9.5
Upcoming Timetable	Remaining classes today for the selected child	D9.5
Recent Results	Preview widget, tagged as a Phase 2 feature preview	D9.5
School Announcements	Count and preview of new notices	D9.5

7.6 Shared Dashboard Behavior

- Live data: all widgets state "Updated just now" on load and tick upward, simulating a refresh every 5 minutes (D9.6).
- Loading state: KPI figures render as a shimmering skeleton before resolving to live values, rather than appearing instantly (D9.6).
- Progress rings and comparison bars animate from 0 to their target value on screen load – used only once per view.
- Role isolation: each dashboard is a fully separate screen and menu structure (D9.7).
- Out-of-scope features appear in navigation only where explicitly tagged Phase 2/3, and are otherwise disabled rather than hidden.

8. Navigation Structure by Role

Sidebar is persistent-collapsible on desktop ($\geq 1280\text{px}$), becomes a temporary drawer on tablet (768–1279px), and is not shown on mobile – the mobile app is used instead for Teacher, Student, and Parent roles.

Tenant Admin

Group	Items
Main	Dashboard, Branches
Academic	Students, Teachers, Timetables, Courses
Finance	Fee & Billing (live) · Payroll, Petty Cash, P&L Reports (Phase 3)
Operations	Leave Management, Resource Logs (live) · HR Management (Phase 2)
Communication	Messages, Appointments, Social Media (Phase 2)
Settings	Tenant Settings, RBAC & Roles, Integrations

Branch Admin

Group	Items
Main	Dashboard
Academic	Students, Teachers, Timetable, Attendance, Personalized Classes (live) · Homework, Results (Phase 2)
Finance	Fee & Billing (live) · Petty Cash (Phase 3)
Operations	Leave Requests, Resource Logs (live) · Appointments, Certificates (Phase 2)
Communication	Messages, Announcements, Social Media (Phase 2)
Calendar	Academic Calendar (Phase 2)

Teacher

Group	Items
Overview	Today's Classes, My Timetable
Classroom	Attendance, Daily Update Log (live) · Homework, Results (Phase 2)
Personal	My Profile, Leave Requests (live) · Salary Slips (Phase 3)

Student & Parent (web)

Role	Items
Student	Home, Timetable, Fees, Digital ID (live) · Homework, Results (Phase 2)
Parent	Home, Attendance, Fees (live) · Homework, Messages, Appointments (Phase 2)

9. Accessibility, Usability & Handoff

9.1 Accessibility Notes

Area	Requirement
Keyboard focus	All interactive elements show a visible focus ring (Royal Blue) — never removed via outline:none
Color contrast	Body text (Dark Slate on White/Light Gray) and button labels meet WCAG AA at minimum
Motion	Animations (skeleton shimmer, ring fill, bar fill) run once on load; no looping decorative motion
Form errors	Always paired with visible text, never color alone, so they remain legible for color-blind users
Reduced motion	Engineering should respect prefers-reduced-motion by disabling ring/bar fill transitions
SMS as accessibility channel	For users with limited data connectivity or older devices, SMS may be the only message that reliably arrives

9.2 Usability Highlights

- Geo-attendance session math (total in-premises time from all IN/OUT/AUTO-OUT/RE-IN stamps) removes the need for teachers to dispute a single dropped connection.
- Combined + per-course invoicing gives parents one clear "total payable" number while preserving itemized detail for admins.
- Phase 2/3 navigation items are shown but disabled with a phase tag rather than hidden — sets accurate expectations during Phase 1 review.
- Skeleton loading states on dashboard KPIs reinforce the real-time claim made in D9.6 of the SOW, rather than a blank card or spinner.
- The parent child-switcher avoids requiring a separate login per child — one guardian account scoped to multiple students.

9.3 Developer Handoff Notes

- Role-based redirect map (Section 6.1) should be implemented at the auth-response layer, not client-side routing guesses.
- Phase-gating: disabled/"Soon"-tagged nav items should be driven by a feature-flag or phase config, not hardcoded per component.
- Session-status logic (5 states) must be computed server-side from the immutable IN/OUT/AUTO-OUT/RE-IN audit log — the UI only renders the resulting state.
- Dashboard "live" refresh (D9.6) is simulated client-side in the prototype for demo purposes; production should poll or subscribe on a 5-minute interval.
- Nepal Pay QR generation must be unique per invoice per billing cycle (D7.8) — tested live in the Milestone 2 walkthrough.

9.4 Prototype Deliverables Referenced in This PRD

File	Covers	Format
tms-auth-prototype.html	Login, Forgot Password, OTP Verification, Reset Password, Two-Factor Authentication	Interactive HTML/CSS/JS, desktop, light mode

File	Covers	Format
tms-dashboards-prototype.html	Tenant Admin, Branch Admin, Teacher, Student, and Parent dashboards	Interactive HTML/CSS/JS, desktop, light mode

9.5 Final Recommendations

- Prioritize a short usability pass with an actual Branch Admin at Damak before Phase 2 kickoff.
- Consider a Nepali-language toggle for parent- and student-facing surfaces in a future phase.
- Keep this document versioned alongside each phase's SOW — Phase 2 will extend, not replace, the personas and IA defined here.

9.6 Change Log

Version	Date	Change
v1.0	7 July 2026	Initial consolidated PRD — Design System, Module 01, and D9 Dashboard
v2.0	7 July 2026	Merged in UX strategy layer: Project Overview, Personas, User Flow, Information Architecture, Wireframe Structure, and expanded Handoff/Recommendations